

# Forecasting Flood Risk One to Three Days Ahead for the Huong River Basin from Open Hydro-Meteorological Data

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**Abstract**—Central Vietnam is the most flood-affected region of the country, yet commune-scale warning products for the Huong River basin remain scarce. This project proposes to forecast river discharge and official flood-alert stage one to three days ahead using only open data: simulated discharge from GloFAS, reanalysis precipitation from ERA5, and archived precipitation forecasts. The work is primarily predictive; prescriptive analytics are explicitly excluded. A 42-year discharge record and an 11-year hourly precipitation grid have already been collected and assembled into a 6,087-record daily table. Preliminary analysis establishes a dominant one-day precipitation–discharge lag ( $r = 0.790$ ) and measures four reference baselines, the strongest of which reaches  $NSE = 0.542$  at a one-day horizon but loses all skill beyond it. The contribution is therefore framed as local correction and translation of an existing global forecast into Vietnam's own alert stages at sub-basin resolution, not as its replacement.

**Keywords**—flood forecasting, streamflow prediction, GloFAS, ERA5, open data, gradient boosting, rare-event classification, Huong River

## I. INTRODUCTION

Central Vietnam is the region of the country most frequently and most severely affected by river flooding. The city of Hue lies on the Huong River, whose catchment of roughly 2,830 km<sup>2</sup> is short and steep: intense rainfall generated by tropical depressions and the northeast monsoon is converted into a flood peak within a single day. Communes along the river and in the adjacent Tam Giang lagoon belt are inundated repeatedly, yet commune-scale warning products remain scarce, and the density of in-situ gauging is limited relative to the damage at stake.

At the same time, global hydro-meteorological datasets have become openly available at a resolution and latency that were unavailable a decade ago. The Global Flood Awareness System (GloFAS) publishes simulated daily river discharge on an approximately 5 km grid. The service documents coverage from 1984, but for this basin the endpoint returns values only from 1997 onward, which we verified directly [1]; we therefore state the record as 1997–2026. The ERA5 reanalysis supplies hourly precipitation and near-surface meteorology over the same period [2]. Both are redistributed without cost and without an access key, which removes the procurement risk that commonly derails student projects.

This report proposes a project that exploits those open datasets to forecast flood risk one to three days ahead for the Huong River basin, and — critically — to express that forecast in the alert-stage vocabulary that Vietnamese authorities already use, rather than in an ad-hoc scale of our own invention.

## II. PROBLEM STATEMENT AND OBJECTIVES

### A. Context of big data

The project is a data-engineering problem before it is a modelling problem. Volume is moderate rather than extreme:

42 years of daily discharge and 11 years of hourly precipitation across 68 grid points amount to approximately 1,500 Parquet files. The difficulty lies in the other dimensions. *Variety* is high: simulated discharge, reanalysis meteorology, archived precipitation forecasts, river-network geometry, a legal threshold document and narrative flood bulletins must all be reconciled onto one daily timeline. *Velocity* is non-trivial: forecast products are re-issued several times per day, so an operational pipeline must ingest on a schedule rather than in a single batch. *Veracity* is the binding constraint: the discharge series is model output, not measurement, and reservoir operation upstream is not represented in it.

### B. Research questions

- **RQ1.** Can daily mean river discharge be predicted 1, 2 and 3 days ahead from gridded precipitation and antecedent discharge, and with what error?
- **RQ2.** With what detection rate and false-alarm ratio can a classifier identify days that exceed an official flood-alert stage?
- **RQ3.** Which sub-basins carry the highest seasonal flood risk?

### C. Analytics approach

Of the four canonical classes of analytics, this project is *primarily predictive*. Table I states the position taken on each class and the reason. The exclusion of prescriptive analytics is deliberate rather than incidental: issuing an evacuation recommendation requires legal authority and accountability that an academic project does not possess, and presenting one would be irresponsible regardless of model quality.

TABLE I  
Position on the Four Classes of Analytics

| Class        | Position     | Reason                                                                                                              |
|--------------|--------------|---------------------------------------------------------------------------------------------------------------------|
| Descriptive  | Supporting   | Seasonality, rainfall–discharge lag structure and historical event reconstruction establish that the signal exists. |
| Diagnostic   | Partial      | Feature attribution explains which driver produced a given warning, but no formal causal design is claimed.         |
| Predictive   | <b>Core</b>  | Both RQ1 and RQ2 are forecasting problems at a 1–3 day horizon.                                                     |
| Prescriptive | Out of scope | Response and evacuation decisions require authority the project does not hold.                                      |

#### D. Objectives

- Assemble a reproducible daily analysis table for the basin from open sources.
- Convert the legal water-level alert stages into discharge thresholds so that the classification target is defined in official terms.
- Establish measured baselines, then train and evaluate forecasting models against them under both idealised and operational input conditions.
- Deliver an interpretable risk product at sub-basin scale, with limitations stated explicitly.

### III. RELATED WORK

GloFAS couples a hydrological model to meteorological forcing to produce global ensemble streamflow forecasts and has been evaluated extensively at continental scale [1]. Its known weakness is precisely our study area: performance degrades in small, steep, monsoon-driven catchments that the global calibration does not resolve well. This motivates local post-processing rather than direct use.

Machine-learning approaches to streamflow prediction have matured substantially. Gradient-boosted decision trees remain strong on tabular hydrological features and train in seconds on datasets of our size [6]. Recurrent architectures have been shown to learn transferable catchment behaviour across large samples [8], although their advantage is clearest where many catchments are pooled, which is not our setting.

Evaluation practice in hydrology is well established. The Nash–Sutcliffe efficiency [3] remains the reference skill score, with the Kling–Gupta efficiency [4] decomposing error into correlation, variability and bias components; published guidance places a satisfactory threshold near 0.5 for streamflow simulation [5]. For rare-event classification, the meteorological contingency measures (probability of detection, false-alarm ratio, critical success index) are preferred to accuracy. Model-agnostic attribution [7] provides the interpretability that operational users require of any warning product.

*Scope note.* This section states the technical foundations the project builds on. A broader review of flood forecasting specific to the Huong–Bo river system is in preparation and

will be consolidated in Report 2.

## IV. DATA REQUIREMENTS AND COLLECTION

### A. Required datasets

Table II lists the six inputs, their role in the model and their current acquisition status. Five are already on disk; the sixth is the project's critical path and is discussed in Section VI.

TABLE II  
Data Requirements and Acquisition Status

| Dataset                                 | Role                          | Span                        | Status      |
|-----------------------------------------|-------------------------------|-----------------------------|-------------|
| River discharge, GloFAS v4 [1]          | Target and lag features       | 1997–2026, daily            | Collected   |
| Precipitation and meteorology, ERA5 [2] | Primary predictors            | 2010–2026, daily and hourly | Collected   |
| Archived precipitation forecasts        | Operational input scenario    | 2022–2026, daily            | Collected   |
| River centre-lines, OpenStreetMap       | Grid-cell to river assignment | Current extract             | Collected   |
| Legal alert stages [9]                  | Defines classification target | Decision 05/2020/QD-TTg     | Verified    |
| Documented flood peaks                  | Calibrates level to discharge | ≥ 15 events, 1999–2026      | In progress |

### B. Grid-cell selection

Because GloFAS represents the river network on a coarse grid, the modelled channel does not coincide exactly with the mapped river. The coordinate initially proposed for the Kim Long gauge returned a long-term mean discharge of only 5.7 m<sup>3</sup>/s, which is physically implausible for this basin. Three independent lines of evidence were combined to select the correct cell: proximity to named river centre-lines obtained from OpenStreetMap; correlation between candidate discharge series, which distinguishes separate rivers from successive reaches of one river; and inversion of long-term mean discharge into an implied catchment area using a regional specific-runoff coefficient.

The third test proved decisive. Cell (16.45°N, 107.50°E) yields a mean discharge of 120.7 m<sup>3</sup>/s, implying a catchment of 2,366–2,943 km<sup>2</sup> against the documented 2,830 km<sup>2</sup> of the Huong basin, an error of 7%. Two larger candidate cells imply catchments of 3,814–4,745 km<sup>2</sup> and 6,144–7,643 km<sup>2</sup> respectively, consistent with positions downstream of successive confluences. The same analysis established that the 5 km grid does not resolve the neighbouring Bo River (938 km<sup>2</sup>) as a separate channel, and the study was therefore narrowed to a single gauging station.

### C. Collection method

Acquisition is implemented as a set of independent tasks, each writing one Parquet file. A task whose output already exists is skipped, so an interrupted run resumes without re-requesting data. Request pacing adapts to the server response: it widens on rate-limit replies and tightens again during clean runs. A lock file prevents concurrent crawler instances.

That last safeguard was added in response to a diagnostic error worth recording. Persistent rate-limit responses were initially attributed to the provider, and a substantial scope reduction was proposed on that basis. The actual cause was three crawler processes running in parallel on the authors' machine and competing with one another. With a single process, the entire collection completed in approximately 35 minutes with no rate-limit response at all, and the proposed scope reduction was withdrawn.

#### D. Processing

Raw files are normalised to Vietnam local time, hourly precipitation is aggregated to daily totals, and validation functions raise an error rather than silently repairing anomalies such as negative discharge, duplicated dates or gaps in the daily index. Gridded precipitation is averaged over three sub-basins (upper, middle and lower) rather than over the whole catchment, preserving the spatial information that upstream rainfall carries for downstream discharge. Lag features, rolling accumulations and an antecedent precipitation index are then derived. All rolling windows are trailing; automated tests assert that no feature reads a value from the future.

### V. PRELIMINARY RESULTS

The proposal is supported by data already collected and processed. The assembled table contains 6,087 daily records and 71 columns spanning 2010-01-01 to 2026-08-31, with no duplicated dates and no missing discharge values.

Mean annual precipitation over the basin is 2,929 mm, consistent with the published climatology of Hue and providing independent support for the specific-runoff assumption used in cell selection. Fig. 1 shows the cross-correlation between basin-mean daily precipitation and discharge: the maximum occurs at a lag of one day ( $r = 0.790$ ), falling to 0.608 at two days and 0.382 at three. Single-day precipitation outperforms 3-, 5- and 7-day accumulations, which is physically consistent with a small, steep catchment and supports the short forecast horizon proposed.

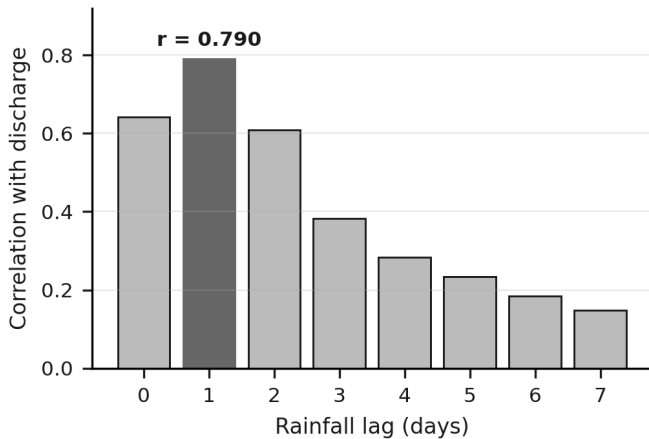


Fig. 1. Cross-correlation between basin-mean precipitation and discharge. The maximum at one day defines the dominant predictive lag.

Seasonality is pronounced (Fig. 2). The months September to December account for 32% of days but 59% of total runoff volume, confirming by measurement the flood season assumed a priori.

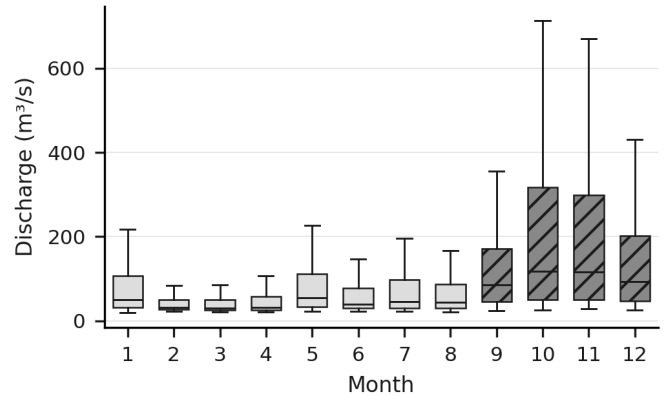


Fig. 2. Monthly distribution of daily discharge. Hatched boxes mark the flood season (September–December).

Four reference models were evaluated on the held-out period from July 2022 (Fig. 3 and Table III). Persistence is the strongest, reaching  $NSE = 0.542$  at a one-day horizon but collapsing to 0.008 at two days and  $-0.198$  at three. This result had an immediate methodological consequence: the acceptance threshold originally set at  $NSE \geq 0.5$  was already met by a trivial baseline and was therefore raised, and differentiated by horizon. The opportunity for a learned model lies at two and three days, where every baseline fails.

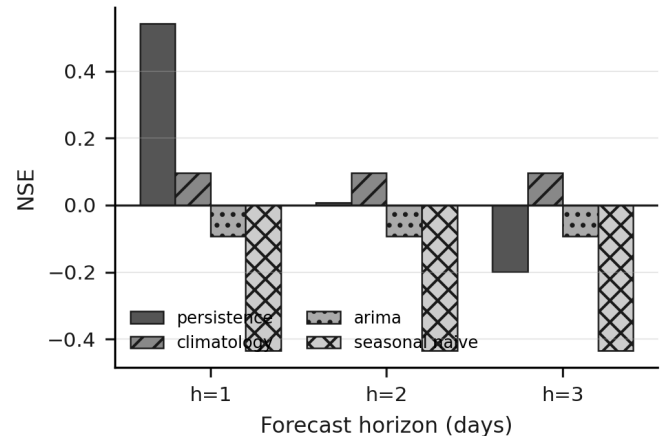


Fig. 3. Baseline skill by forecast horizon. All reference models lose predictive skill beyond one day.

TABLE III  
Baseline Nash–Sutcliffe Efficiency, Test Period 2022–2026

| Model                    | h=1    | h=2    | h=3    |
|--------------------------|--------|--------|--------|
| Persistence              | 0.542  | 0.008  | −0.198 |
| Climatology              | 0.097  | 0.097  | 0.096  |
| ARIMA(2,1,2) on log flow | −0.093 | −0.093 | −0.093 |
| Seasonal naive           | −0.434 | −0.434 | −0.435 |

A scheduled job has additionally issued and archived a fresh seven-day forecast every morning since the project began. This

forward record is the only available route to a like-for-like comparison against the raw global forecast, because no archive of past GloFAS discharge forecasts is published; the historical series is model analysis, not forecasts issued in advance. This constraint was discovered during implementation and required revising the evaluation design.

## VI. TIMELINE AND RISK MANAGEMENT

The project runs over a nine-week term with four graded submissions. Report 1 (this document, 10%) is due in week 2; Report 2, covering data and exploratory analysis (20%), in week 4; Report 3, covering modelling and evaluation (40%), in week 8; and a consolidated final report (10%) in week 9, followed by an individually assessed oral examination (20%).

Data acquisition, originally scheduled across the first three weeks, completed within week 1. The critical path is now the compilation of documented flood peaks required to convert alert stages into discharge thresholds. A request for observed stage–discharge records was not viable, leaving event-based calibration as the sole route; if fewer than ten usable events are obtained, the fallback is a percentile-based risk scale, which must then be renamed so that it is not presented as the official alert system.

Two further risks are recorded. First, the anomaly of 14 June 2025, where a discharge of 2,055 m<sup>3</sup>/s coincides with basin-mean precipitation of 0.9 mm, suggests reservoir release; the basin contains the Ta Trach and Binh Dien reservoirs, whose operation GloFAS does not represent. Second, model inputs during training are reanalysis precipitation whereas operational inputs are forecast precipitation, so results will be reported under both an idealised and an operational scenario, and the difference between them treated as a result in its own right.

## VII. CONCLUSION

This project does not seek to replace an existing global flood forecast. It aims to correct that forecast locally for the Huong River basin and to translate it into the alert stages that Vietnamese authorities act upon, at sub-basin resolution and with interpretable attribution. The data required has been collected and verified, the predictive signal has been measured, and reference baselines have been established, so the modelling work proceeds against a quantified bar rather than an assumed one.

Three limitations are stated at the outset rather than deferred: the discharge series is simulated and not measured; the grid resolution does not separate the neighbouring Bo River, restricting the study to one gauging station; and the product is academic work, not an official warning service.

## . REFERENCES

- [1] L. Alfieri, P. Burek, E. Dutra, B. Krzeminski, D. Muraro, J. Thielen, and F. Pappenberger, “GloFAS – global ensemble streamflow forecasting and flood early warning,” *Hydrol. Earth Syst. Sci.*, vol. 17, no. 3, pp. 1161–1175, 2013.
- [2] H. Hersbach *et al.*, “The ERA5 global reanalysis,” *Q. J. R. Meteorol. Soc.*, vol. 146, no. 730, pp. 1999–2049, 2020.

- [3] J. E. Nash and J. V. Sutcliffe, “River flow forecasting through conceptual models part I – a discussion of principles,” *J. Hydrol.*, vol. 10, no. 3, pp. 282–290, 1970.
- [4] H. V. Gupta, H. Kling, K. K. Yilmaz, and G. F. Martinez, “Decomposition of the mean squared error and NSE performance criteria: implications for improving hydrological modelling,” *J. Hydrol.*, vol. 377, no. 1–2, pp. 80–91, 2009.
- [5] D. N. Moriasi, J. G. Arnold, M. W. Van Liew, R. L. Bingner, R. D. Harmel, and T. L. Veith, “Model evaluation guidelines for systematic quantification of accuracy in watershed simulations,” *Trans. ASABE*, vol. 50, no. 3, pp. 885–900, 2007.
- [6] G. Ke *et al.*, “LightGBM: a highly efficient gradient boosting decision tree,” in *Proc. Adv. Neural Inf. Process. Syst. (NeurIPS)*, 2017, pp. 3146–3154.
- [7] S. M. Lundberg and S.-I. Lee, “A unified approach to interpreting model predictions,” in *Proc. Adv. Neural Inf. Process. Syst. (NeurIPS)*, 2017, pp. 4765–4774.
- [8] F. Kratzert, D. Klotz, G. Shalev, G. Klambauer, S. Hochreiter, and G. Nearing, “Towards learning universal, regional, and local hydrological behaviors via machine learning applied to large-sample datasets,” *Hydrol. Earth Syst. Sci.*, vol. 23, no. 12, pp. 5089–5110, 2019.
- [9] Prime Minister of Vietnam, *Decision No. 05/2020/QĐ-TTg on water levels corresponding to flood alert levels on rivers nationwide*, Hanoi, 31 Jan. 2020.
- [10] Open-Meteo, “Free open-source weather API.” [Online]. Available: <https://open-meteo.com>. Data licensed under CC BY 4.0. [Accessed: 17 Sep. 2026].